INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

RADIAC DETECTOR DT-236/PDR-75

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for construction and testing of the Radiac Detector DT-236/PDR-75, individual, indirect reading, neutron-gamma dosimeter for detecting, measuring neutron and gamma doses received by individual soldiers operating in nuclear environments.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-11268 - Parts, Materials, and Processes Used in Electronic Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey, 07703-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC - N/A FSC 6665

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MIL-M-13231 - Marking of Electronic Items
MIL-F-14072 - Finishes for Ground Electronic Equipment

STANDARDS

MILITARY

MIL-STD-252 - Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-414 - Sampling Procedure and Table for Inspection by Variables for Percent Defective
MIL-STD-454 - Standard General, Requirements for Electronic Equipment
MIL-STD-781 - Reliability Design Qualification and Production Acceptance Tests; Exponential Distribution
MIL-STD-810 - Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue,Philadelphia, PA.)

2.1.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

DL-C5085569 - Radiac Detector DT-236/PDR-75
NBS Handbook #63 and #75-Washington, D.C. 2024

#63 Protection Against Neutron Radiation up to 30 Million Electron Volts

#75 Measurement of Absorbed dose of Neutrons and Mixtures of Neutrons and Gamma Rays

Title 15, Chapter II, Part 200, Section 200.115 of the Code of Federal Regulations

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A3160505 - Nuclear Survivability Requirements for Detector, Radiac DT-236/PD and Computer Indicator CP-696()/UD(U) 18 March 1974
TB9-6665-286-35 - Calibration procedure for Radiac Set AN/PDR-75

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exception has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract, the contractor shall furnish sample units for first article inspection and approval (see 4.3.1).

3.2 Construction. The Radiac Detector DT-236/PDR-75 shall be constructed in accordance with the requirements of this specification and DL C5085569. The dosimeter shall consist of a phosphate glass gamma sensitive element and a neutron sensitive PIN diode. These elements shall be encased in a container in the shape and approximate size of a wrist watch.

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply (see 4.4).

3.3.1 Recycled and reclaimed materials. Recycled or reclaimed materials in lieu of virgin materials shall be used to the maximum extent possible without jeopardizing the specified performance and reliability of the radiac detector.

3.3.2 PIN diode neutron detecting element. The neutron detecting PIN diode shall be a Harshaw DN 156 device or equivalent.

3.3.3 Phosphate glass gamma detecting element. The phosphate glass gamma detector shall be Toshiba Glass Company FD-7 or equivalent.

3.3.4 Activation materials. Effort shall be made to reduce unnecessary amounts of materials in the instruments which have a thermal neutron activation cross section. These materials include but are not

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limited to gold, silver, copper, vanadium, fluorine, manganese, cobalt, germanium, mercury, iodine, bromine, antimony, selenium, boron, and cadmium.

3.3.5 Tamper resistant. The design of the dosimeter shall be such that it will be very difficult to create an artificial dose reading by manipulation or tampering.

3.3.6 Environmental sealing. The dosimeter shall be sufficiently enclosed to prevent damage to the radiation sensitive elements when the dosimeter is exposed to the environmental conditions of 4.10. Material used for enclosing the dosimeter shall not cause changes in the phosphate glass or neutron diode or radiation detection capabilities.

3.3.7 Surface. The surface of the dosimeter shall be as smooth as possible.

3.3.8 Strap. Provisions shall be made to allow the dosimeter to be worn on the wrist and also to allow a serial number to be stamped/engraved on the dosimeter.

3.3.9 Calibration of the AN/PDR-75 field reader. The AN/PDR-75 field reader shall be calibrated in accordance with the calibration procedure described in Department of the Army Technical Bulletin TB9-6665-286-35. The reference locket set in the Maintenance Kit MK-2512/PDR-75 shall be calibrated every 6 months by the US Army Test Measurement and Diagnostic Equipment (TMDE) Support Group during lifetime of the contract (see 4.1.2).

3.4 Finish. The DT-236/PDR-75 shall exhibit a surface finish characteristic of a standard plastic molding process. (see 4.4)

3.5 Marking. Marking shall be in accordance with MIL-M-13231 (see 4.4).

3.5.1 Serial numbers. Each DT-236/PDR-75 shall be serial numbered (see 4.4).

3.6 Interchangeability. Like units, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.6).

3.7 Size and weight.

3.7.1 Size. The size of the DT-236/PDR-75 shall not exceed 0.88 cubic inches (16 cubic cm).

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3.7.2 Weight. The maximum total weight of the DT-236/PDR-75 shall not exceed 70 gms.

3.8 Safety engineering.

3.8.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection (see 4.8).

3.8.2 Corner and edge rounding. All exposed corners and edges shall be rounded to eliminate possible injury to personnel due to lacerations and cuts.

3.8.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.9 Performance requirements.

3.9.1 Response. The dosimeter shall respond from one to 1000 cGy of gamma (or x-ray) and neutron radiation either separately or in combination (i.e., mixed radiation fields such as from a reactor or nuclear weapon) (see 4.9).

3.9.2 Accuracy. When measuring radiation in environmental conditions as specified in paragraphs 3.10 to 3.10.12 with detailed interface conditions given in paragraphs 3.9 to 3.9.14, the dosimeter, shall record total neutron and gamma doses that read on an AN/PDR-75 calibrated field reader with a 95 percent confidence, within ±30 percent or ±30 cGy whichever is greater of the true total dose at that location except for recorded total neutron and gamma doses made at +68°F (+20°C) ±7°F (±3.9°C) where the accuracy shall be within ±20 percent or ±20 cGy, whichever is greater of the true total dose. The required accuracy applies to the total dose from representative spectra which may be encountered as a result of nuclear weapons employment. Representative spectra for a mixed gamma-neutron initial radiation field and a gamma only residual field are given in 4.9.

3.9.3 Gamma accuracy. The total integrated gamma dose accuracy will be determined as described in 3.9.2, 3.9.7, 4.9.1 and 3.9.1.

3.9.4 Neutron accuracy. The total integrated neutron dose accuracy will be determined as described in 3.9.2, 3.9.8, 4.9.2 and 3.9.1.

3.9.5 Predose reading. Undosed dosimeters shall be read on a

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calibrated AN/PDR-75 field reader. The neutron an gamma channel readings shall be 0±5 cGy and 0±5 cGy, respectively (see 4.9.13).

3.9.6 Precision. A statistically valid sample of 20-25 dosimeters shall be subjected to a calibration run at mid-scale reading (500 cGy). The standard deviation of the dose indication between dosimeters of the sample shall not exceed 5 percent of the group mean dose indication (see 3.9.1, 4.9.4 and 4.9.5).

3.9.7 Gamma energy dependence. The dosimeters shall be capable of recording, at mid-scale (500 cGy) in free air at ambient temperature, tissue cGy dose of gamma and x-ray above 80 KeV and across the energy range as given in 4.9.1.2 to within ±20 percent at the 95 percent confidence level. For energy below 80 KeV, the response of the dosimeter shall exhibit a continual decrease in readings for equal doses of lower energy, such that the readings shall decrease not more than 50 percent at 50 KeV and no more than 5 percent at 35 KeV. The 95 percent confidence level assumes a normal distribution of errors (see 4.9.1).

3.9.8 Neutron energy dependence. The dosimeters shall be capable of recording the total neutron dose in tissue cGy for the spectra of neutron energies up to 14 MeV (see 4.9.2).

3.9.9 Fast neutron response. The dosimeters shall be capable of recording in free air at ambient temperature the total fast neutron dose in tissue cGy from a free air pulsed nuclear reactor providing dose-rates not less than 10^7 cGy/sec with a 75 micro-sec pulse or less measured at full width at half maximum within ±20 percent of the true tissue dose at the 95 percent confidence level from the neutron environment described in 4.9.3. and 4.9.2.

3.9.10 Thermal neutron sensitivity. The thermal neutron sensitivity shall be determined and be documented as soon as feasible (see 4.9.6). The thermal neutron tissue cGy is defined as follows:

$$\text{One tissue cGy} = 3.2 \times 10^9 \text{ thermal neutron/cm}^2$$

3.9.11 Rate dependence. The dosimeter recording capability for gamma, neutron, or mixed gamma-neutron fields of radiation shall be independent of the intensity of radiation up to at least 10^7 cGy/sec for neutrons and 10^8 cGy/sec for gammas (to preclude ultra high dose rate saturation) (see 4.9.7).

3.9.12 Linearity. The deviation from linearity using the zero and 620 cGy reading as a datum point, shall not be more than ±10% at 250 and 750 cGy and shall not be more than ±10% at full scale reading (999 cGy) for each dosimeter with 95 percent confidence level (4.9.8 and 4.9.9).

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a. Three sample groups of dosimeters shall be selected at random from each month's production. A fourth sample group shall be selected quarterly. Sample size shall be based on production lot size in accordance with:

Lots size	Sample size
3 to 8	3
9 to 15	3
16 to 25	3
26 to 40	3
41 to 65	3
66 to 110	3
111 to 180	3
181 to 300	3
301 to 500	4
501 to 800	5
801 to 1300	7
1301 to 3200	10
3201 to 8000	15
8001 to 22000	20
22001 to 110000	25
110001 to 550000	25
550001 and over	25

b. Initial readings prior to dose exposure shall be taken for all units.

c. One sample group shall be dosed to each of the following dose levels; 250, 500 and 750 cGy (tissue). The fourth group shall be dosed to 950 cGy (tissue) quarterly beginning with the first month. (see 4.9.8 and 4.9.9)

d. All gamma dosing for any one test sequence shall utilize the same calibrated Cobalt-60 or Cesium-137 gamma source with calibration by a secondary standard traceable to National Standard. The specified doses traceable to the National Standard shall be delivered to within ±5 percent. Neutron dosing shall utilize a calibrated neutron source, delivering a neutron energy spectrum comparable with the Aberdeen fast burst reactor, with calibration by a secondary standard traceable to the National Standard. The specified doses shall be delivered to within ±10 percent. The systematic uncertainties in the dosimetery used for determining actual doses of gamma or neutron delivered may be considered as not affecting the linearity test result.

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e. The uncertainties in the determination of the control limits (denoted as e) shall be calculated as the square root of the sum of squares of the following terms: 1) the population standard deviation of the dosimetry at "500" level, 2) the population standard deviation of the dosimetry at the level for which limits are being calculated (this may have the same value as 1a above), and 3) the sample standard deviation of the "500" level readings of the DT236 under test (calculated per example B3 of MIL-STD-414).

f. The mean and sample standard deviation (with n-1 degrees of freedom) at each dose level shall be calculated in accordance with Example B3 of MIL-STD-414.

g. The ten percent control limits shall be established at 10 percent above;

$$\frac{\text{(Dose given)}}{\text{(Dose given at "500" level)}} \times (\text{(Mean reading at "500")} + 1.96e)$$

and at ten percent below;

$$\frac{\text{(Dose given)}}{\text{(Dose given at "500" level)}} \times (\text{(Mean reading at "500")} - 1.96e)$$

for the nominal 250 and 750 readings respectively.

h. The control limits shall be used as the specification upper and lower limits (U and L of MIL-STD-414) and compared against the distribution of actual sample readings at the 250 and 750 levels in accordance with MIL-STD-414, double specification limit, variability unknown, standard deviation method. Use table below to obtain the value for M.

Sample size	M
3	18.86
4	16.45
5	14.39
7	12.20
10	10.54
15	9.46
20	8.92
25	8.63

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i. Similar methodology shall be utilized quarterly (every third lot) beginning with the first lot, for the nominal 950 cGy (tissue) dose level. Control limits are equal to:

$$U=\frac{(\text{Dose given at 950})\times((\text{Mean reading at "500"})+1.96e)}{(\text{Dose given at 500})}+10\%,$$

$$\text{and } L=\frac{(\text{Dose given at 950})\times((\text{Mean reading at "500"})-1.96e)}{(\text{Dose given at 500})}-10\%.$$

NOTE: All references to MIL-STD-414 herein shall be read as the issue of MIL-STD-414 of 11th June 1957.

Example calculation

This sample illustrates the method of calculation of test limits and acceptance for the "250" level only. The calculation to be performed at the other level follows exactly the same pattern with the appropriate data values substituted.

Data: at "500"

Received dose: 500 cGy standard deviation of dosimetry at 500 (d_{500}): 2%

Readings: 506, 496, 499, 502, 497, 510, 493, 488, 487, 501, 504, 492, 500, 493, 495, 496, 508, 508, 489, 499, 493, 502, 495, 498, 495 cGy.

Mean reading (X_{500}): 497.84 cGy (Calculated per MIL-STD-414)
Sample standard deviation(s_{500}): 6.256 cGy (1.257%) (Calculated per MIL-STD-414)

Data: at "250"

Received dose: 230 cGy Standard deviation of dosimetry at 250 (d_{250}): 2.1%

Readings: 233, 227, 235, 232, 229, 224, 234, 226, 226, 222, 226, 237, 232, 232, 229, 224, 229, 234, 230, 236, 222, 235, 223, 225, 230 cGy.

Mean reading(X_{250}): 229.28 cGy (Calculated per MIL-STD-414)

Sample standard deviation (S_{250}): 4.605 cGy (2.008%) (Calculated per MIL-STD-414).

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Calculation of e_{250}:

$$e_{250} = \sqrt{(d250)^2+(d500)^2+(s500)^2}$$

$$= \sqrt{(2.1^2+2^2+1.257^2)} \text{ percent}$$

$$= \sqrt{(4.41+4.0+1.5800)} \text{ percent}$$

$$= \sqrt{9.9900 \text{ percent}} = 3.1607 \text{ percent}$$

Calculation of upper control limit (U):

$$\frac{\text{Dose at "250"}}{\text{Dose at "500"}} = \frac{230}{500} = 0.46$$

$$U_{250} = \frac{\text{(Dose "250")}}{\text{(Dose "500")}} \times \text{(Mean reading at "500")} + 1.96e + 10 \text{ percent}$$

$$= 0.46 \times (497.84 + 1.96 \times 3.1607\%) \times 1.1$$

$$= 0.46 \times (497.84 \times 1.06195) \times 1.1$$

$$= 267.5 \text{ cGy}$$

Calculation of lower control limit (L):

$$\frac{\text{Dose at "250"}}{\text{Dose at "500"}} = 0.46$$

$$L_{250} = \frac{\text{(Dose "250")}}{\text{(Dose "500")}} \times \text{(Mean reading at "500")} - 1.96e - 10\%$$

$$= 0.46 \times 497.84 - 1.96 \times 3.1607\% \times 0.9$$
$$= 0.46 \times 497.84 \times (1-06195) \times 0.9$$
$$= 193.4 \text{ cGy}$$

Assessment of result (see also MIL-STD-414 Example B3)

$$Q_1 = (X_{250}-L_{250})/S_{250}$$

$$= (229.28-193.4)/4.605$$

$$= 7.79$$

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From table B-5 P^1 = 0 percent

$Qu = (U_{250}-X_{250})/S_{250}$

$= (267.5-229.28)/4.605$

$= 8.30$

From table B-5 Pu = 0 percent

P = Pl + Pu = 0 percent

From table B-3 M = 8.63 percent

As P is less than M the linearity test is passed at the 250 cGy level.

3.9.13 Fade. When not exposed to ionizing radiation, each dosimeter prior dosed shall not change its reading by more than 30 percent for the period of 24 hours to 1 year after irradiation and stored at 25°C with a 95 percent confidence level (see 4.9.10).

3.9.14 Isotropism. Sensitivity and accuracy shall be affected as little as possible by the geometric orientation of the dosimeter. In no case shall the reading vary by more than ±15 percent on a 4 geometry basis when the dosimeter is rotated in all possible orientations to the incident radiation (see 4.9.11 and 4.9.12). Based on a successful first article test, isotropism testing will not be required during production.

3.10 Service conditions. The DT-236/PDR-75 shall be subjected to and meet the service conditions below. For the purpose of service conditions, specified performance is defined as meeting the following requirements:

Requirement paragraph	Inspection paragraph	Measurement
3.9.3	4.9.1	Gamma accuracy
3.9.4	4.9.2	Neutron accuracy
3.9.6	4.9.4	Precision, gamma
3.9.6	4.9.5	Precision, neutron

3.10.1 High temperature. The DT-236/PDR-75 shall meet the specified performance at 125°F (52°C), the storage temperature shall be 160°F (+71°C) (see 4.10.1).

3.10.1.1 Ambient Temperature. The DT-236/PDR-75 shall meet the specified performance at 68°F (20°C) ±7°F (±3.9°C) (see 4.10.1.1).

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3.10.2 Low temperature. The DT-236/PDR-75 shall meet the specified performance at -25°F (-32°C) and the storage temperature shall be -70°F (-57°C) (see 4.10.2).

3.10.3 Humidity. The DT-236/PDR-75 shall exhibit no physical damage such as corrosion, rust, blistering, swelling, or deterioration of parts and materials and shall meet the specified performance when subjected to 94±4 percent relative humidity. The dosimeter shall meet specified performance without degradation after completion of the test (see 4.10.3).

3.10.4 Altitude. The DT-236/PDR-75 shall meet the specified performance at 10,000 ft (3048m) above sea level. The dosimeter shall meet specified performance without degradation after completion of the test (see 4.10.4).

3.10.5 Temperature shock. The DT-236/PDR-75 shall be capable of withstanding temperature shocks without degradation and shall meet the specified performance between -65°F (-54°C) and +145°F (+63°C) (see 4.10.5).

3.10.6 Immersion. The DT-236/PDR-75 shall show no evidence of leakage after being tested in accordance with 4.10.6.

3.10.7 Vibration. The DT-236/PDR-75 shall operate without degradation in specified performance and shall not suffer any mechanical damage after being subjected to the vibration tests specified in 4.10.7.

3.10.8 Shock. The DT-236/PDR-75 shall operate without degradation in specified performance and shall suffer no mechanical damage after being subjected to the shock test specified in 4.10.8.

3.10.9 Dust. The DT-236/PDR-75 shall operate without degradation in specified performance after being subjected to the dust test of 4.10.9.

3.10.10 Salt fog. After the salt fog test of 4.10.10, the DT-236/PDR-75 shall operate without degradation in specified performance and shall show no evidence of degradation such as flaking, pitting, blistering or loosening of finish or metal surface exfoliation (see 4.10.10).

3.10.11 Fungus. After the fungus test of 4.10.11, the DT-236/PDR-75 shall be visually examined. The DT-236/PDR-75 shall show no more than sparse microbial growth with restricted tubular growth development in an area 10 percent or less of the total area and no more than 6 unrelated minute colonies with mycelial development in areas only in other than

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critical circuit portions; such as terminal spacing, with sparse growth due to random contamination or traces of unmixed material ingredients. In addition, the DT-236/PDR-75 shall operate without degradation in specified performance after exposure to the test conditions of 4.10.11. Materials used in the manufacture, fabrication, construction and treatment of the DT-236/PDR-75 shall comply with Requirement 4 of MIL-STD-454.

3.10.12 Nuclear survivability. The DT-236/PDR-75 shall meet the specified performance after being subjected to the nuclear environment of 4.10.12.

3.11 Reliability. The DT-236/PDR-75 shall have a lower test (0^1) mean-time-between-failure (MTBF) of 16,800 hours while operating under the following conditions (see 4.11):

a. Operating temperature: $-25^{\circ}F$ to $+125^{\circ}F$ ($-35^{\circ}C$ to $+52^{\circ}C$)

b. Environment: Ground mobile

c. Vibration: $2.2g \pm 0.22g$

d. Duty cycle: Continuous operation

3.12 Workmanship. The DT-236/PDR-75 shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of MIL-P-11268 (see 4.12 and 4.4).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements

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in this specification shall not relieve the contractor of the responsibility of assuring that all products and supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.2 Calibration of field reader AN/PDR-75. The contractor shall perform calibration as described in TB9-6665-286-35 of the field reader AN/PDR-75, as required in paragraph 3.3.9 prior to reading DT-236 dosimeters before and after the conduction of test specified in section 4 of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3); does not include packaging.

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspection of equipment before packaging (see 4.5).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish 210 first article units of the DT-236/PDR-75.

4.3.2 First article inspection. The first article inspection shall consist of the inspection specified in Table I, and shall be performed in the order specified.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract.

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TABLE I. First article inspection.

Inspections	Reqt para	Test para	Order of tests 2/					
			30 units	30 units	30 units	30 units	30 units	60 units
1. Inspections covered by subsidiary documents 1/	3.3 3.4 3.5 3.12	4.4 4.4 4.4 4.12	Inspection to be performed on all units					
2. Group A inspection 1/	See table II		Inspection to be performed on all units					
3. Group B inspection	See table III		Each inspection to be performed on 26 units for a total of 208 units					
4. Group C inspection								
Ambient temperature	3.10.1.1	4.10.1.1	1					
High temperature	3.10.1	4.10.1	3					
Low temperature	3.10.2	4.10.2	2					
Humidity 3/	3.10.3	4.10.3	5					
Immersion	3.10.6	4.10.6			1			
Dust 3/ 5/	3.10.9	4.10.9		2				
Vibration 3/	3.10.7	4.10.7			2			
Altitude 5/	3.10.4	4.10.4		1				
Shock 3/	3.10.8	4.10.8			3			
Salt fog 3/ 5/	3.10.10	4.10.10		3				
Fungus 3/ 5/	3.10.11	4.10.11			4			
Temperature shock	3.10.5	4.10.5	4					
Neutron accuracy	3.9.4	4.9.2				1		
Fast neutron response	3.9.9	4.9.2				2		
Thermal neutron sensitivity	3.9.10	4.9.6				3		
Rate dependence	3.9.11	4.9.7				4		

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TABLE I. First article inspection. (cond't)

Inspections	Reqt para	Test para	Order of tests 2/ 30 units	30 units	30 units	30 units	30 units	60 units
Precision, neutron	3.9.6	4.9.5				5		
Linearity, neutron	3.9.12	4.9.9				6		
5. Unique group of test								
Gamma energy dependence 4/	3.9.7	4.9.1					1	
Neutron energy dependence 4/	3.9.8	4.9.2					2	
Fade	3.9.13	4.9.10					3	(5 units)
Isotropism neutron	3.9.14	4.9.12					4	
Isotropism gamma	3.9.14	4.9.11					(5 units) 5	
Nuclear survivability 4/	3.10.12	4.10.12						
Electromagnetic pulse	3.10.12	4.10.12.1					7	
Nuclear air blast	3.10.12	4.10.12.2					6	
Thermal radiation	3.10.12	4.10.12.3					8	
6. Group D inspection								
Reliability	3.11	4.11						1

1/ The inspection 1 and 2, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements.

2/ The numbers in the unit columns in the table, covered by inspections 4 and 5, specify the order of inspections for the indicated units. The 3 nuclear survivability tests will be performed after tests 1 thru 5. Thermal radiation test will follow the electromagnetic pulse test.

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3/ Equipment must be thoroughly washed, cleaned, dried and refurbished, if necessary before proceeding with subsequent tests.

4/ Ten units shall be submitted in order shown, to gamma energy dependence and neutron energy dependence tests. Ten other units shall be submitted in order shown, to nuclear survivability tests.

5/ The dust, altitude, salt fog and fungus tests shall be performed only during first article testing.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirements specified in the contract:

Item	Where required
Parts, materials, and processes; general	3.3
Finish, protective	3.4
Marking	3.5
Workmanship	3.12

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections in table II. Lots shall be formed from units that pass this inspection. Each lot shall be subjected to sampling inspection in accordance with table III. Lots in which samples exhibit any failures shall be screened for that failure prior to being subjected to group B inspection.

4.5.1.1 Order of inspection with group A. The group A inspection shall be performed in an order satisfactory to the government.

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TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test Paragraph
Visual and mechanical	3.2 3.8 3.12	4.4 4.8 4.12
Predose reading	3.9.5	4.9.13

TABLE III - Sampling plan for visual and mechanical inspection and predose reading

Lots Size	Sample Size
2 to 8	1/
9 to 15	13
16 to 25	13
26 to 50	13
51 to 90	13
91 to 150	13
151 to 280	13
281 to 500	20
501 to 1200	34
1201 to 3200	42
3201 to 10,000	50
10,001 to 35,000	60
35,000 to 150,000	74
150,001 to 500,000	90
500,001 to over	102

NOTES:

1. 1/ Indicates entire lot must be inspected.
2. Acceptance number in all cases is zero.

4.5.2 Group B inspection. Group B inspection shall be performed on lots that have passed group A inspection. This inspection shall consist of the inspection listed in table IV. Each lot shall be subjected to sampling inspection in accordance with table V and table VI. Lots in which samples exhibit any failures shall be screened for that failure prior to the units within the lot being subject to group C inspection.

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TABLE IV. Group B inspection.

Inspection	Requirement para	Test para
Response	3.9.1	4.9.3
Accuracy	3.9.2	4.9.3, 4.9.1
Gamma accuracy	3.9.3	4.9.1
Linearity, gamma	3.9.12	4.9.8
Precision, gamma	3.9.6	4.9.4
Isotropism, gamma 1/	3.9.14	4.9.11
Interchangeability	3.6	4.6
Size and weight	3.7	4.7

1/ Not required during production if successfully demonstrated during first article inspection.

TABLE V - Sampling plan for response-accuracy-gamma accuracy linearity, gamma-precision, gamma-isotropism, gamma-size and weight

Lots Size	Sample Size
2 to 8	3
9 to 15	3
16 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1200	15
1201 to 3200	18
3201 to 10,000	22
10,001 and over	29

NOTE: Acceptance number in all cases is zero.

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TABLE VI - Sampling plan for interchangeability

Lots Size	Sample Size
2 to 8	2
9 to 15	2
16 to 25	3
26 to 50	5
51 to 90	5
91 to 150	6
151 to 280	7
281 to 500	9
501 to 1200	11
1201 to 3200	13
3201 and over	15

NOTE: Acceptance number in all cases is zero.

4.5.2.1 Order of inspection within group B. Group B inspection shall be performed in an order satisfactory to the government.

4.5.3 Group C inspection. Group C inspection shall be performed on units in lots that have passed group A and group B inspections. The inspection shall consist of the inspections specified in table VII. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for group C inspection. ninety samples of the DI-236/PDR-75 shall be selected at random. Thirty samples for each subgroup in table VII from every 25,000 units or fraction thereof greater than or equal to 12,500 units. If the fraction is less than 12,500, no group C inspections shall be required. The samples shall be selected from the first quality-conformance lot consisting of units from each 25,000 units or fraction thereof produced.

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TABLE VII. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
High temperature	3.10.1	4.10.1
Low temperature	3.10.2	4.10.2
Temperature shock	3.10.5	4.10.5
Humidity 1/	3.10.3	4.10.3
Subgroup 2		
Vibration 1/	3.10.7	4.10.7
Shock 1/	3.10.8	4.10.8
Immersion	3.10.6	4.10.6
Subgroup 3 2/		
Neutron accuracy	3.9.4	4.9.2
Fast neutron response	3.9.9	4.9.3
Thermal neutron sensitivity	3.9.10	4.9.6
Rate dependence	3.9.11	4.9.7
Precision, neutron	3.9.6	4.9.5
Linearity, neutron	3.9.12	4.9.9

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after their inspection before proceeding with subsequent inspection.

2/ The subgroup 3 tests shall be performed only once at midpoint of each year production quantities. Terminal neutron sensitivity shall be performed only once in middle of production quantities.

4.5.3.2 Group C failures. Action required relative to group C failures shall be as specified in the contract.

4.5.3.3 Reinspection of conforming group C sample units. Unless otherwise specified, undosed sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group B and group A inspection, in that order.

4.5.4 Group D inspection. This inspection shall consist of the tests specified in table VIII and shall be performed on units from lots which have been subjected to and met group A and group B inspection.

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4.5.4.1 Sampling for group D inspection. Sixty samples shall be selected at random from the first quality conformance lot consisting of units from each 25,000 units or fraction thereof produced. If the quantity of the last fraction produced consist of less than 12,500 units, the selection of group D samples for the preceding test shall include units from the remaining fraction.

TABLE VIII. Group D inspection.

Inspection	Requirement paragraph	Test paragraph
Reliability	3.11	4.11

4.5.4.2 Group D failures. In case of group D failure occurrence, procedures and requirements specified elsewhere in the contract shall be followed.

4.5.4.3 Reinspection of group D sample units. Unless otherwise specified, sample units which have been subjected to, and have passed group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

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4.7 Size and weight.

4.7.1 Size. The size of the DT-236/PDR-75 shall be measured. It shall meet the requirement of 3.7.1.

4.7.2 Weight. The DT-236/PDR-75 shall be weighed. The weight shall not exceed that given in 3.7.2.

4.8 Safety engineering inspection. A visual inspection shall be performed in the presence of a government safety representative to determine that all requirements of 3.8 have been incorporated in the equipment. Inability to meet these requirements shall constitute a failure of the test. Results of this inspection shall be included in the Safety Assessment Report, if required in the contract.

4.9 Performance tests.

4.9.1 Gamma radiation.

4.9.1.1 Gamma accuracy calibration. The x-ray machine and gamma sources of $137C_s$ and $60C_o$ used for test shall be calibrated with regard to delivered dose by a secondary standard. The secondary standard shall be a Victoreen R-meter "or equivalent" calibrated at the National Institute of Standards and Technology (NIST) "or equivalent" within a 12 month period prior to use. For gamma with energies up to 12 Mev the associated calibration dosimetry shall utilize thermoluminescence powder in sealed glass containers having energy compensating shields and proper electron equilibrium layer thickness of polyethylene (see 3.9.3).

4.9.1.1.1 Gamma reading. Five gamma reading shall be taken when reading each separate gamma dosed DT-236. The readings shall be recorded and averaged.

4.9.1.2 Gamma energy dependence testing. The dosimeters, in free air and at ambient temperature shall be subjected to x-rays and gammas of various energies up to 12 Mev. For production of the desired x-ray energy spectrum the contractor shall utilize a constant potential machine having x-ray filters described in Title 15, Chapter II, part 200, Section 200.115 of the Code of Federal Regulations. The stated accelerating voltages are 50, 100,150, 200 and 250 KVCP. The approximate effective energies are 40, 70, 120, 170 and 215 Kev. Gamma sources of $137C_s$ and $60C_o$ will provide medium energies of 0.66 and 1.3 Mev. Exposures shall be made at selected energies to determine the dosimeter's fractional response to a given dose by comparison to that measured by an acceptable standard at the dosimeter location. Prior to use, the contractor shall

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obtain government approval of the acceptable standard to be used. Having determined the dosimeter's gamma dose fractional response values, the contractor shall apply, by integration, the measured responses to the representative gamma environment given in 4.9.3 to determine the total integrated gamma dose error. The combined total integrated neutron plus gamma dose error as derived from the dosimeter's fractional dose response versus energy and the representative nuclear spectrum shall not exceed that stated in 3.9.2.

4.9.2 Neutron radiation.

4.9.2.1 Neutron accuracy calibration. The contractor shall utilize the most accurate method when determining the delivered neutron dose at the location of the test item with calibration traceable to NIST. Calibrated tissue equivalent ion chambers, and associated particle counting dosimetry is recommended. The first collision neutron dose table for tissue in NBS Handbook #75 shall be the dose conversion reference standard.

4.9.2.1.1 Neutron readings. Fifteen neutron readings shall be taken when reading each separate neutron dosed DT-236. The first ten readings shall be discarded and the remaining five recorded and averaged.

4.9.2.2 Neutron energy dependence testing. The dosimeters in free air and at ambient temperature shall be subjected to various neutron energies from thermal to 14 Mev. The T (P, N) He3 reaction can provide mono-energetic neutrons for measurements at 0.5, 1.0 and 2.0 Mev. A fast burst reactor will provide degraded watt spectrum neutrons with an average energy of 1.8 Mev. Mono-energetic neutrons of 2.5 and 14 Mev are available from D (d, n) He3 and T (d, n) He4 reactions. Exposure shall be made at selected energies to determine the dosimeter's fractional response to a given dose by comparison to that dose measured by an acceptable standard at the dosimeter location. Prior to use, the contractor shall submit to the government for approval an acceptable standard. The contractor shall compute the total integrated neutron dose error using the measured fractional dose response versus energy values and the representative neutron spectra given in 4.9.3. The combined total integrated neutron plus gamma dose error as derived from the dosimeter's fractional dose response and representative nuclear spectra shall not exceed that stated in 3.9.2.

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4.9.3 Representative neutron and gamma ray spectra and ratios.

a. Neutron spectrums.

	Output fraction	
Energy (MeV)	Fission	Thermonuclear
12.2 - 15	1.568(-4)	7.06(-2) read as 7.06×10^{-2}
10.0 - 12.2	8.932(-4)	2.56(-2)
8.19 - 10.0	3.480(-3)	1.41(-2)
6.36 - 8.19	1.392(-2)	1.47(-2)
4.97 - 6.36	3.457(-2)	1.80(-2)
4.07 - 4.97	3.507(-2)	1.70(-2)
3.01 - 4.07	1.072(-1)	2.60(-2)
2.46 - 3.01	8.898(-2)	1.90(-2)
2.35 - 2.46	2.323(-2)	5.00(-3)
1.83 - 2.35	1.203(-1)	2.80(-2)
1.11 - 1.83	2.181(-1)	6.20(-2)
0.55 - 1.11	1.983(-1)	8.50(-2)
0.11 - 0.55	1.403(-1)	1.02(-1)
0.0335 - 0.110	1.550(-2)	3.65(-1)
5.83(-4) - 3.35(-2)	0	1.22(-1)
1.01(-4) - 5.83(-4)	0	2.40(-2)
2.90(-5) - 1.01(-4)	0	2.00(-3)
2.90(-5)	0	0

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b. Initial gamma-ray spectrum.

Energy (Mev)	Output Fraction
6.0 - 8.0	1.77(-4) read as $1.77 \times 10^{(-4)}$
5.0 - 6.0	4.5(-3)
4.0 - 5.0	1.5(-2)
3.0 - 4.0	4.23(-2)
2.0 - 3.0	1.3(-1)
1.5 - 2.0	1.37(-1)
1.0 - 1.5	2.24(-1)
0.9 - 1.0	5.46(-2)
0.8 - 0.9	5.7(-2)
0.7 - 0.8	5.05(-2)
0.6 - 0.7	7.25(-2)
0.5 - 0.6	9.0(-2)
0.4 - 0.5	5.7(-2)
0.3 - 0.4	5.46(-2)
0.2 - 0.3	1.2(-2)
0.1 - 0.2	3.0(-4)

c. Fission product spectrum.

Energy (Mev)	Fraction
2.5 - 5.0	4.5(-2) read as 4.5×10^{-2}
1.5 - 2.5	2.4(-1)
.5 - 1.5	7.15(-1)

d. Neutron/gamma ratio.

Small yield, air burst, fission spectrum.

Dose-rad	n/γ
100	2
3000	4

Large yield, air burst, thermonuclear spectrum.

Dose-rad	n/γ
100	1/6
3000	1/2

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e. Neutron capture and/or induced gamma-ray spectrum. Spectrum depends on atmospheric and soil composition. The energy range may extend to 12 MeV, higher energy (7-12 MeV) fraction depending on nitrogen in air and quantity of iron and aluminum in soil.

f. Tissue dose conversion values are found in NBS Handbook #63 and #75.

4.9.4 Precision gamma. A minimum of 10 dosimeters shall be exposed at ambient temperature to either $60C_o$ or $137C_s$ for gamma dose to approximately 500 cGy (tissue). Between 46 and 48 hours after dosing the gamma dose indication from each dosimeter shall be determined by reading it on a calibrated AN/PDR-75 field reader 15 times and average and record the last 5 readings. The standard deviation of the gamma dose indication between dosimeters of the sample shall not exceed 5 percent of the group mean gamma dose indication. (see 3.9.6 and 4.9.1.1).

4.9.5 Precision neutron. A minimum of 10 dosimeters shall be exposed at ambient temperature on a fast burst reactor for neutron dose to approximately 500 cGy neutron. Between 46 and 48 hours after dosing, the neutron dose indication from each dosimeter shall be determined by reading it on a calibrated AN/PDR-75 field reader 15 times and average and record the last 5 readings. The standard deviation of the neutron dose indication between dosimeters of the sample shall not exceed 5 percent of the group mean neutron dose indication. (see 3.9.6 and 4.9.2.1).

4.9.6 Thermal neutron sensitivity. The dosimeter shall be exposed at ambient temperature to a thermalized fast burst reactor environment to a level of at least 100 cGy. Dose response of the neutron and gamma detectors shall be documented (see 3.9.10).

4.9.7 Rate dependence. The dosimeters shall be exposed at ambient temperature on a fast burst reactor for neutron dose (see 3.9.9) and flash x-ray machine for gamma dose to a level of 500 cGy neutron and 500 cGy gamma. The readings accuracy shall be in accordance with 3.9.2 (see 3.9.11).

4.9.8 Linearity gamma. The dosimeters shall be exposed to either a $60C_o$ or $137C_s$ source for gamma dose to approximately 250, 500, 750, and 950 cGy gamma dose. The accuracy limits described in 3.9.12 shall apply.

4.9.9 Linearity neutron. The dosimeters shall be exposed on a fast burst reactor for neutron dose to approximately 250, 500, 750 and 950 cGy of neutron dose.

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4.9.10 Fade. The dosimeters shall be exposed to approximately 800 cGy neutron and 800 cGy gamma on a fast burst reactor and either $60C_o$ or $137C^s$. A reading will be taken 48 hours ± 1 hour after exposures. Subsequent readings will be taken over a period of 3 months. The resulting curves shall be extrapolated on an exponential base to approximate the fade for a period of 1 year and shall meet the requirement of 3.9.13.

4.9.11 Isotropism gamma. The dosimeters at ambient temperature shall be rotated in 45° increments about an axis normal to the plane of the dosimeter through 360° and exposed to 100 cGy gamma in each position. In a like manner, the dosimeter shall be rotated in 45° increments about an axis through the centerline of the dosimeter, parallel to the plane of the dosimeter through 360° and exposed to 100 cGy gamma (see 3.9.14).

4.9.12 Isotropism, neutron. The dosimeters shall be rotated in 45° increments about an axis normal to the plane of the dosimeter through 360° and exposed to 100 cGy neutron in each position. In a like manner, the dosimeters shall be rotated in 45° increments about an axis through the centerline of the dosimeter, parallel to the plane of the dosimeter through 360° and exposed to 100 cGy neutron (see 3.9.14).

4.9.13 Predose reading. Undosed dosimeters shall be read on a calibrated AN/PDR-75 field reader. The neutron and gamma channel readings shall be read separately and results recorded (see 3.9.5). Five reading shall be taken and averaged.

4.10 Service conditions. The service condition tests of paragraph 3.10 shall be performed as detailed below. Each DT-236/PDR-75 subjected to these tests shall have passed group A and B tests and shall be resubmitted to and pass the visual and mechanical group A tests after completion of all service condition tests. Where operation of the test item is required during any service condition test, the test item shall be required to pass the specified performance.

4.10.1 High temperature. The DT-236/PDR-75 shall be subjected to the test of Method 501.1, Procedure II, of MIL-STD-810. The storage temperature (Step 4) shall be 160^oF (71^oC). In Step 7, the operating temperature shall be $+125^oF$ ($+52^oC$) (see 3.10.1).

4.10.1.1 Ambient temperature. The DT-236/PDR-75 shall be subjected to the test methods defined in the inspection paragraphs contained in paragraph 3.10.1.1. The ambient temperature shall be 68^oF ($+20^oC$) $\pm 7^oF$ ($\pm 3.9^oC$).

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4.10.2 Low temperature. The DT-236/PDR-75 shall be subjected to the test of Method 502.1, Procedure I, of MIL-STD-810. The storage temperature (Step 2) shall be -70°F (-57°C) and shall be maintained for a period of not less than 2 hours following stabilization of the test item. Low operating temperature (Step 4) shall be -25°F (-32°C) (see 3.10.2).

4.10.3 Humidity. The DT-236/PDR-75 shall be subjected to the test of Method 507.1 of MIL-STD-810, Procedure II. In Step 6 of procedure, measurements shall be taken during the last 5 hours of the last cycle (see 3.10.3).

4.10.4 Altitude. The DT-236/PDR-75 shall be subjected to the test of Method 500.1, Procedure I of MIL-STD-810. The equipment shall be heldat 50,000 feet (15,240m) non-operating for 1 hour. The operating altitude in Step 2 shall be 10,000 feet (3,048m) (see 3.10.4).

4.10.5 Temperature shock. The DT-236/PDR-75 shall be subjected to test Method 503.1, Procedure I, of MIL-STD-810. The high temperature exposure during this test shall be 145°F (63°C) instead of the specified 160°F (71°C). The low temperature exposure shall be -65°F (-54°C) (see 3.10.5). Separate gamma and neutron doses shall be recorded before and after test.

4.10.6 Immersion. The DT-236/PDR-75 shall be subjected to the test of Method 512.1, Procedure I, of MIL-STD-810 (see 3.10.6).

4.10.7 Vibration. The DT-236/PDR-75 shall be subjected to the test of Method 514.2, Procedure VIII, Curve V of MIL-STD-810 (see 3.10.7).

4.10.8 Shock. The DT-236/PDR-75 shall be subjected to the test of Method 516.2, Procedure II of MIL-STD-810 (see 3.10.8).

4.10.9 Dust. The DT-236/PDR-75 shall be subjected to the test of Method 510.1, Procedure I, of MIL-STD-810 (see 3.10.9).

4.10.10 Salt fog. The DT-236/PDR-75 shall be subjected to the test of Method 509.1, Procedure 1, of MIL-STD-810 (see 3.10.10).

4.10.11 Fungus. The DT-236/PDR-75 as prepared for operation, shall be subjected to the test of Method 508.1, Procedure I of MIL-STD-810. There shall be abundant growth colonization on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the dosimeter is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the dosimeter (see 3.10.11).

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4.10.12 Nuclear survivability. The DT-236/PDR-75 shall be subjected to the nuclear environment as specified in A3160505 (see 3.10.12).

4.10.12.1 Electromagnetic pulse test. The dosimeter shall be placed within a long wire electromagnetic pulse facility or its equivalent. Test data to satisfy the EMP requirements in A3160505 may be assembled by extrapolation.

4.10.12.2 Nuclear air blast test. The dosimeter shall be placed in a nuclear air blast simulation facility and exposed to the air blast environment as specified in A3160505. The dosimeter shall meet the specified performance herein following this test.

4.10.12.3 Thermal radiation test. The dosimeter shall be exposed to the thermal radiation levels specified in A3160505. Since the available simulation facilities provide these thermal radiation levels only over relatively small areas, many individual exposures will be required so that each of the exposed surfaces will receive the required thermal exposure. The dosimeter following exposures, shall show no degradation and shall meet specified performance herein.

4.11 Reliability testing.

4.11.1 Reliability demonstration test. The reliability of the DT-236/PDR-75 dosimeter unit shall be demonstrated in accordance with Test Plan IVC of MIL-STD-781 using a specified MTBF (O_o) of 33,600 hours and lower tests MTBF (O_1) of 16,800 hours. Sample size shall be 60 units. Since the device is a passive device, it may be considered as always being "on."

4.11.2 Parameter to be measured. The specified performance for the test items will be conducted during the ambient temperature cycle. Each dosimeter will be exposed to 400 cGy gamma and 400 cGy neutron before reliability testing commences. Each day, a reader will read the dosimeters and drift (fade) data for the dosimeter recorded.

4.11.3 Test conditions. Test environment for the reliability test shall be under the following conditions:

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a. Temperature and on-off cycle. The item shall be continuously operated under the following temperature cycling:

Transit	Stability	Transit	Stability	Transit	Stability
Ambient to -25°F (-32°C)	-25°F (-32°C)	-25°F (-32°C) to +125°F (+52°C)	+125°F (+52°C)	+125°F (+52°C) to Ambient	Ambient
1 hour	9 hours	2 hours	9 hours	1 hour	2 hours

b. Vibration. Vibrate at 2.2 ± 0.22g peak acceleration at any non-resonant frequency between 20 to 60 Hz measured at the mounting points of the unit. Total vibration time in a 24 hour period shall be at least 4 hours (2 hours at low temperature).

4.11.4 Failure definition. A failure is defined as the inability of the equipment to perform its required function. This includes catastrophic failures as well as any event which causes departure or deviation from performance parameters required by this specification. Failure confirmation, categorizing, analysis and action shall all be in accordance with MIL-STD-781. Proper instrumentation shall be provided to insure immediate recognition of failure.

4.12 Visual and mechanical inspection. These inspections shall be performed in any order which is satisfactory to the government. The units shall be examined for the applicable defects listed in MIL-STD-252 (see 3.8).

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of of protection shall be as specified by the acquisition activity.

6. NOTES. (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Radiac Detector DT-236/PDR-75 is intended for detecting and measuring combined neutron and gamma tissue dose in centiGray from nuclear detonations or from fallout.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

c. Marking and shipping of samples.

d. Place of final inspection.

e. Number of first articles to be submitted for approval (see 3.1).

6.3 Subject terms (key word) list.

Nuclear environments
Measurements of neutron and gamma doses
Computer-Indicator Radiac CP-696/PDR-75
Radiac Set AN/PDR-75

This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQL's/LTPD's (Acceptable Quality Levels/Lot Tolerance Percent Defective) from military specifications.

Custodian:
Army - CR

Preparing activity:
Army - CR
Project 6665-A510

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-49421A	2. DOCUMENT DATE (YYMMDD) 5 June 1990

3. DOCUMENT TITLE: RADIAC DETECTOR DT-236/PDR-75

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME U.S. ARMY CECOM	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code) AMSEL-ED-TO FORT MONMOUTH, NJ 07703-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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